

SQL Data Types

Unit I · Lecture 4 · MySQL (SQL/PL-SQL)

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UNIT I

LECTURE 4

SQL DATA TYPES





RECAP

Previous Lecture Recap

→ SQL Command Categories

DDL, DML, DCL, TCL — each serves a distinct role in managing data.

→ DDL Commands

CREATE, ALTER, DROP — used to define and modify database structures.

→ DML Commands

INSERT, UPDATE, DELETE, SELECT — used to manipulate data records.

Learning Objectives

By the end of this lecture, you will be able to:

1

Define Data Types

Understand what SQL data types are and why they matter.

2

Classify Data Types

Identify Numeric, String, Date/Time, Boolean, and Binary types.

3

Compare Similar Types

Distinguish CHAR vs VARCHAR, INT vs FLOAT vs DECIMAL.

4

Apply Best Practices

Choose the right data type for real-world scenarios.



FOUNDATION

What is a Data Type?

A **data type** tells the database *what kind of value* a column will store — a number, text, date, or image.

Accuracy

Prevents storing wrong values (e.g., text in an age field).

Efficiency

Right types save storage and speed up queries.

Integrity

Enforces consistent, valid data across all rows.

Classification of SQL Data Types



Numeric

INT, FLOAT, DECIMAL, DOUBLE



String

CHAR, VARCHAR, TEXT



Date & Time

DATE, TIME, DATETIME, TIMESTAMP



Boolean & Binary

BOOLEAN, BIT, BLOB

Every column in a SQL table must be assigned a data type at the time of creation.

Numeric Data Types

Type	Storage	Range / Precision	Real-World Use
INT	4 bytes	−2.1B to 2.1B	Student ID, Employee ID
SMALLINT	2 bytes	−32,768 to 32,767	Age, Class Strength
BIGINT	8 bytes	Very large whole numbers	Bank account balance, Population
DECIMAL	Variable	Exact precision (e.g., 10,2)	Salary, Product price
FLOAT / DOUBLE	4 / 8 bytes	Approximate decimal values	Scientific readings, GPS coords

❏ Use **DECIMAL** for money — never FLOAT, as FLOAT can introduce rounding errors in financial calculations.

STRING TYPES

Character / String Data Types



CHAR(*n*)

Fixed-length string. Always stores exactly *n* characters. Unused space is padded. Best for data of known, constant length.

Example: Gender (M/F), Country Code (IN)



VARCHAR(*n*)

Variable-length string up to *n* characters. Uses only the space needed. Ideal for names, emails, addresses.

Example: Student Name, Email ID



TEXT

Stores **large blocks of text** — up to 65,535 characters. Used when content length is unpredictable.

Example: Product descriptions, Comments, Feedback

Date & Time Data Types

Type	Format	Range	Example Use
DATE	YYYY-MM-DD	1000-01-01 to 9999-12-31	Date of Birth, Joining Date
TIME	HH:MM:SS	−838:59:59 to 838:59:59	Class Start Time, Shift Hours
DATETIME	YYYY-MM-DD HH:MM:SS	1000 to 9999	Order Placed At, Event Schedule
TIMESTAMP	YYYY-MM-DD HH:MM:SS	1970 to 2038 (UTC)	Record Created/Updated At
YEAR	YYYY	1901 to 2155	Batch Year, Academic Year

TIMESTAMP automatically updates when a record is modified — ideal for audit trails.

Boolean and Binary Data Types

BOOLEAN

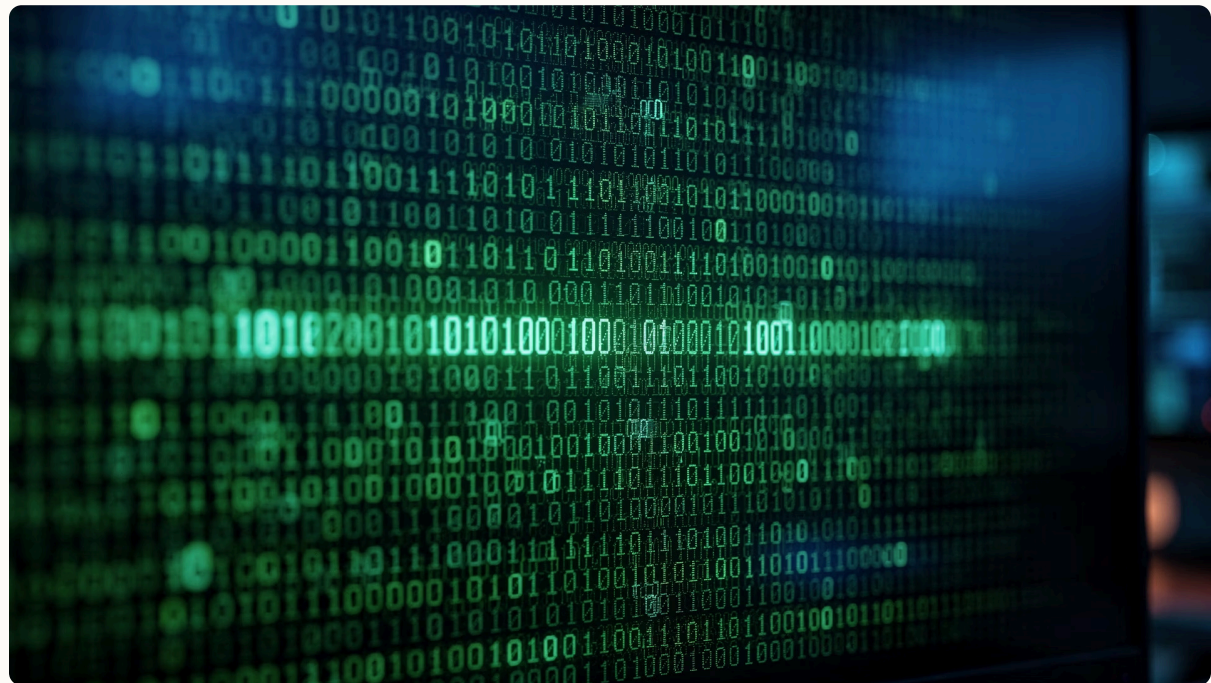
Stores **TRUE** or **FALSE** (internally 0 or 1). Used for flags: Is_Active, Is_Verified, Has_Paid.

BIT(n)

Stores **bit-field values** (binary digits). BIT(1) is often used as a boolean alternative for compactness.

BLOB

Binary Large Object — stores images, audio, video, or documents as raw binary data. Variants: TINYBLOB, BLOB, MEDIUMBLOB, LONGBLOB.



COMPARISON

CHAR vs VARCHAR

Feature	CHAR	VARCHAR
Length	Fixed	Variable
Storage	Always n bytes	Actual length + 1–2 bytes
Speed	Faster retrieval	Slightly slower
Padding	Pads with spaces	No padding
Best For	Fixed codes (PIN, Gender)	Names, Emails, Addresses

Quick Rule of Thumb

If the value **always** has the same length → use **CHAR**.

If the value **varies** in length → use **VARCHAR**.

Example: Mobile number stored as CHAR(10) is more efficient than VARCHAR(10) since it is always 10 digits.

COMPARISON

INT vs FLOAT vs DECIMAL

INT

Whole numbers only. No decimals.
Fast and memory-efficient.

Use for: IDs, counts, quantities, age.

Example: student_id INT

FLOAT / DOUBLE

Approximate decimals. Stores very large or very small values. May have rounding errors.

Use for: Scientific data, GPS, measurements.

Example: temperature FLOAT

DECIMAL(p, s)

Exact decimals. p = total digits, s = digits after decimal. No rounding errors.

Use for: Salary, prices, financial data.

Example: salary DECIMAL(10,2)

❏ For financial applications, always prefer **DECIMAL** over FLOAT — precision matters when money is involved.

BEST PRACTICES

Choosing the Correct Data Type

01

Understand the Data

Is it a number, text, date, or binary? Know what values will be stored.

02

Consider the Range

Will values exceed INT range? Use BIGINT. Need decimals? Choose DECIMAL or FLOAT.

03

Minimize Storage

Don't use BIGINT when SMALLINT suffices. Smaller types = faster queries.

04

Avoid Over-Engineering

Don't use TEXT for short names. VARCHAR(100) is sufficient for most name fields.



REAL-WORLD EXAMPLES

Data Types in the Real World



Student Database

student_id INT, name VARCHAR(100), dob DATE, gpa
DECIMAL(3,2)



Employee Database

emp_id BIGINT, department VARCHAR(50), salary
DECIMAL(10,2), joining_date DATE



Banking System

account_no BIGINT, balance DECIMAL(15,2), last_txn
TIMESTAMP, is_active BOOLEAN



E-Commerce Platform

product_id INT, description TEXT, price DECIMAL(8,2),
image BLOB

Data Type Size & Storage Considerations

A diagram showing the storage size for various SQL data types. Each data type is listed on the left, and its corresponding size in bytes is listed on the right, connected by a horizontal line. The data types are: TINYINT (1 byte), SMALLINT (2 bytes), INT (4 bytes), BIGINT (8 bytes), FLOAT (4 bytes), DOUBLE (8 bytes), CHAR(n) (n bytes), VARCHAR(n) (n+1 or n+2 bytes), DATE (3 bytes), DATETIME (8 bytes), and TIMESTAMP (4 bytes).

TINYINT	1 byte
SMALLINT	2 bytes
INT	4 bytes
BIGINT	8 bytes
FLOAT	4 bytes
DOUBLE	8 bytes
CHAR(n)	n bytes
VARCHAR(n)	n+1 or n+2 bytes
DATE	3 bytes
DATETIME	8 bytes
TIMESTAMP	4 bytes

Why Storage Size Matters

Choosing a smaller type reduces disk usage and speeds up table scans — especially important for tables with millions of rows.

- **Millions of rows** × even 4 extra bytes = significant wasted space
- Smaller columns fit more rows in a single disk page
- Always choose the **smallest type** that safely holds your data

Common Mistakes When Choosing Data Types

Using FLOAT for Money

FLOAT introduces rounding errors. Always use **DECIMAL** for currency and financial fields.

Using TEXT for Short Values

TEXT is heavy. Use VARCHAR for names, emails, and short descriptions.

Oversizing VARCHAR

VARCHAR(1000) for a name field wastes allocation metadata. Use VARCHAR(100) or less.

Storing Dates as Strings

Using VARCHAR('2024-01-15') breaks date functions and sorting. Always use **DATE** or **DATETIME**.

Using BIGINT Unnecessarily

BIGINT uses 8 bytes. Use INT (4 bytes) or SMALLINT (2 bytes) when the range allows it.

CONCEPTUAL EXAMPLE

Sample Student Table — Data Types Applied

Column Name	Data Type	Size	Reason
student_id	INT	4 bytes	Whole number, no decimals needed
student_name	VARCHAR(100)	Variable	Names vary in length
gender	CHAR(1)	1 byte	Always single character: M or F
date_of_birth	DATE	3 bytes	Standard date, no time needed
gpa	DECIMAL(4,2)	Variable	Exact decimal (e.g., 8.75)
is_active	BOOLEAN	1 byte	Active/Inactive flag (TRUE/FALSE)
enrolled_at	TIMESTAMP	4 bytes	Auto-tracks when record was created

Data Type Selection Activity

Choose the **most appropriate data type** for each field below. Discuss with your neighbor!

#	Field	Your Answer	Hint
1	Product price in an online store	?	Needs exact decimal precision
2	Student's mobile number	?	Always 10 digits, no arithmetic
3	Date an order was placed	?	Needs both date and time
4	Employee photo	?	Binary file storage
5	Number of items in stock	?	Non-negative whole number



Answers: 1→DECIMAL, 2→CHAR(10), 3→DATETIME, 4→BLOB, 5→INT or SMALLINT

DISCUSSION

Classroom Discussion

Design a Student Database

Work in pairs. Define a **Student** table with at least **8 fields**. For each field, choose the data type and justify your choice.

Think About

What information does a university store about each student?

Decide the Type

Is each field numeric, text, date, or binary?
Fixed or variable length?

Justify Your Choice

Why is your chosen type better than alternatives?
Consider storage and accuracy.



Quiz — Test Your Understanding

1

Which data type is best for storing a student's salary after graduation?

a) INT b) FLOAT c) **DECIMAL** ✓ d) TEXT

2

CHAR(10) vs VARCHAR(10) — which uses less storage for 'BCA'?

a) CHAR b) **VARCHAR** ✓ c) Both same d) Depends

3

Which type auto-updates when a row is modified?

a) DATE b) DATETIME c) **TIMESTAMP** ✓ d) YEAR

4

What does BLOB store?

a) Long text b) **Binary data (images, files)** ✓ c) Dates d) Boolean flags

5

Which numeric type stores only whole numbers?

a) **INT** ✓ b) FLOAT c) DECIMAL d) DOUBLE



Assignment: Create a conceptual Employee table with 10 fields, specifying data types and justifying each choice. Submit next lecture.

Key Takeaways & Next Lecture Preview

Key Takeaways

- Data types define what kind of value a column can hold
- Use **DECIMAL** for money, **INT** for IDs, **VARCHAR** for names
- CHAR is fixed-length; VARCHAR is variable — choose wisely
- TIMESTAMP auto-tracks record changes; ideal for audit logs
- Always choose the **smallest type** that fits the data range

Coming Up Next — Lecture 5

→ SQL Operators

Arithmetic, Comparison, Logical, and Special operators in SQL.

→ SQL Expressions

Building meaningful conditions using operators in WHERE clauses.

→ Hands-On Practice

Writing queries using operators on a real student table.